



Harnessing Phase Change Materials for Off Grid Refrigeration

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Abstract

The use of phase change materials (PCMs) for off-grid cooling in rural areas is investigated. The PCM PlusICE A4 is selected because of its melting point and latent heat. PCM packs are distributed inside the fridge cabin. During power outage, adequate cooling is obtained for 6 hours. PCM fridges are more ecological and less expensive than PV-driven systems with a battery but exhibit lower performance. Their use is contingent on substantial solar energy availability. This isn't always the case in the target regions.

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1 Introduction

Electricity has evolved into an indispensable daily necessity. Ensuring affordable and reliable access to it is crucial. Unfortunately, this is not consistently the reality in the Global South, where millions of people currently live without dependable access to electricity. A lack of electricity is very problematic for any situation that requires refrigeration. In fact, it is reported that bad storage conditions are one of the main factors behind food wastage in Africa. The addition of these areas to the state grid system is, most of the time, difficult to implement.

As conventional refrigeration methods often rely on continuous power sources, they face limitations in off-grid scenarios where electricity access is scarce or unreliable. PCM integration tackles this issue by storing and releasing thermal energy during the phase transition process, providing a sustainable and reliable alternative for refrigeration. This report investigates the science behind PCM material selection and its compatibility with refrigeration systems. It also explores the potential benefits in terms of energy efficiency, reduced environmental impact as well as the economic feasibility of this solution.

This report also aims to contribute significant insights into the feasibility of PCM technology in off-grid refrigeration. The goal being to push towards more sustainable cooling methods that fall in line with the United Nations' Sustainable Development Goals (SDGs).

2 PCM Fridges

PCMs are materials that go through their solid-liquid transformation at a temperature that falls within a specific operating range for a certain thermal application. A PCM absorbs energy from its surroundings as it changes phase. The energy absorbed is known as the PCM's latent heat of fusion. A material critically remains at a constant temperature during the transition. This particular property is what makes PCMs stand out as a potential solution to the initial problem. Indeed, a constant temperature is the ideal situation for a fridge. PCMs are also lightweight, portable and only depend on the chosen material's properties. They don't rely on external factors. (Fleischer 2015)

2.1 PCM Selection

PCMs can be divided into three different categories: organics, inorganics, and eutectics which are a combination of the latter two. The most important property is the PCM's melting point or range as it must fall around the desired temperature for the fridge to operate at. The second selection criterion is the latent heat of the PCM. The higher the latent heat, the more energy our PCM can absorb and the less PCM quantity we need. But the downside of a high latent energy is a long PCM recharge time, when electricity is available. The recharge time is the time the PCM needs to resolidify by exchanging heat with the outer environment. (Fleischer 2015)

As a freezer option was not included, PCMs with a subzero melting point were not considered. The target range was between 2°C and 5°C. To choose which PCM must be used according to the desired temperature of service, several ones were tested with a calorimeter. Inorganic PCMs tests indicated a big phase transition temperature range so they were not used. Organic PCMs were tested and the PCM PlusICE A4 was selected. It has a melting temperature of 4°C, which is suitable for this case. As can be seen on the right side of Figure 1, with this PCM, the temperature is maintained around 4°C during power outage until all the PCM melts. This PCM is also inexpensive and safe, therefore well suited for home applications.(Riffat et al. 2022)

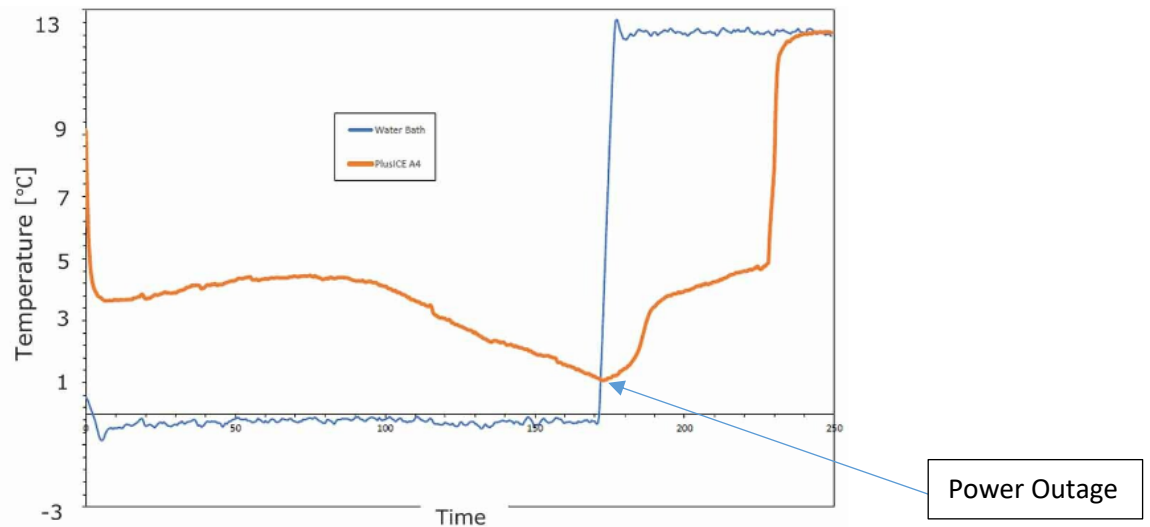


Figure 1 Thermal Performance of PCM PlusICE A4 and Water Bath(Riffat et al. 2022)

2.2 Phase Change Material Integration in Refrigeration Systems

Considering the target population of this technology, a photovoltaic (PV) powered refrigeration system was designed to act as the on-grid power source. PCMs intervene during nighttime or low sunlight conditions. The integration of PCMs revolves around their precise placement within the refrigerator. PCMs are encapsulated in packs that are then distributed inside the fridge. Specifically, thermal storage packs are positioned in important locations of the unit, with a favored placement in the refrigerator cabins. There is a PCM mass constraint of 3kg for spacing reasons because when dealing with high quantities of PCM, it becomes difficult to channel all the heat to the PCM. This is due to the low heat conductivity of PCMs. Looking at Table 1, 3kg would secure around 620kJ of the 14400kJ consumed daily by a regular fridge. This is not enough. By optimizing fridge components like the compressor, and making them more adapted to high heat environment, acceptable off-grid cooling can be guaranteed for approximately 6 hours. The modifications made to the fridge components won't be covered by this report.(Radhi, Johain, and Khalifa 2021)

The system consists of three compartments: the PV collector, the refrigeration cycle and the refrigeration cabinet. Figure 2 provides a visual representation of the functioning mechanism. The PV collector generates direct current (DC) to power the compressor. The DC compressor controller regulates the compressor speed based on solar irradiance. At the same time, the refrigeration unit cools the cabinet with variable cooling loads. Inside the cabinet, PCM packages facilitate heat exchange with the air during the day.(Riffat et al. 2022)

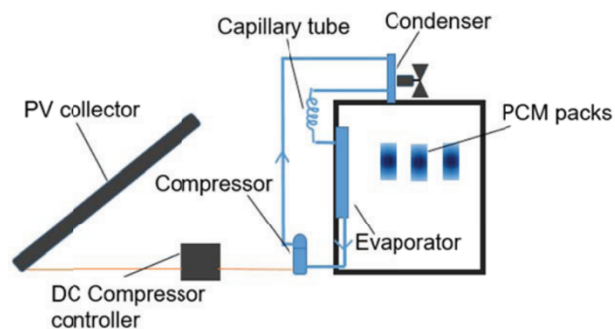


Figure 2 System Model (Riffat et al. 2022)

During daytime, at the beginning of a cycle, the fridge functions like a traditional fridge. The PCMs inside the packs cool and solidify and exchange heat with the exterior environment. When the PV compartment stops providing current to the refrigerator, PCMs start absorbing the heat coming into the fridge, turning liquid. When sunlight intensity is back up, typically the next day, the cycle restarts. (Khan, Saidur, and Al-Sulaiman 2017)

When tested, the temperature inside the fridge fluctuates around the PCM's melting point until all of it melts. Then, the fridge air temperature rises steadily until the power is back. In contrast, fridges that use a battery show more stable temperatures until the battery runs out of power.

2.3 Cost and Environmental Impact Analysis

2.3.1 Environmental Impact Analysis

The ecological impact of this refrigerator depends on the environmental effects of both the photovoltaic-based system and the integrated PCM. The impact of transportation has been neglected from the assessment due to the inherent challenges in accurately calculating it. Notably, for our evaluation of the PCM PlusICE A4, we reference the values of Octadecane, a substance studied in the ecological analysis. The Environmental Cost Indicator (ECI) is utilized to assess the overall environmental impact of this product as seen in figure 3.

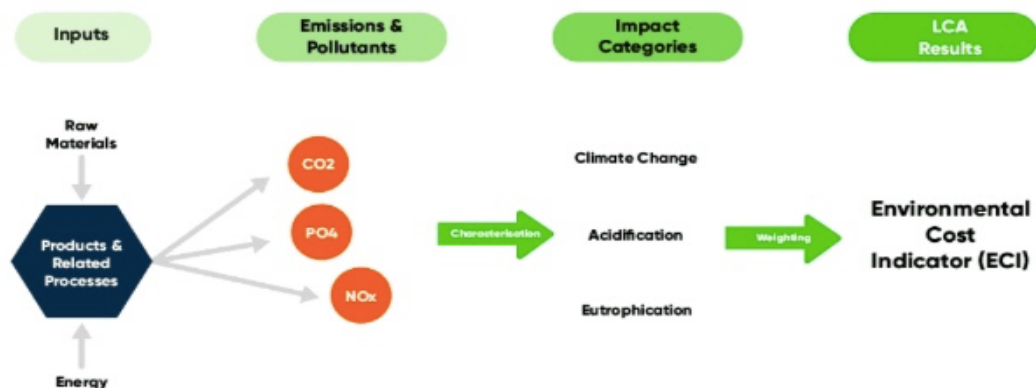


Figure 3 Explanation of ECI (Luc Hillege 2023)

ECI encompasses all emissions associated with a process or product, providing insights into its environmental production impact and assigning a monetary cost to “repair” the inflicted damage. This includes its life cycle, emissions, pollution, and the resource depletion associated with such a product.

The ECI values for different PCMs were assessed based on their ability to release or store 100kJ of energy. For Octadecane, this value is 5.47€ per 100kJ (Aridi and Yehya 2022). The environmental impact related to PCMs is calculated at 32.9€. In comparison, using a lithium-ion battery to store the same amount of energy would cost 242€ in ECI. This indicates that using PCMs is a much more ecological solution for domestic use. This suggests that PCMs have a low environmental footprint, are non-polluting during their life service and hold recycling potential.

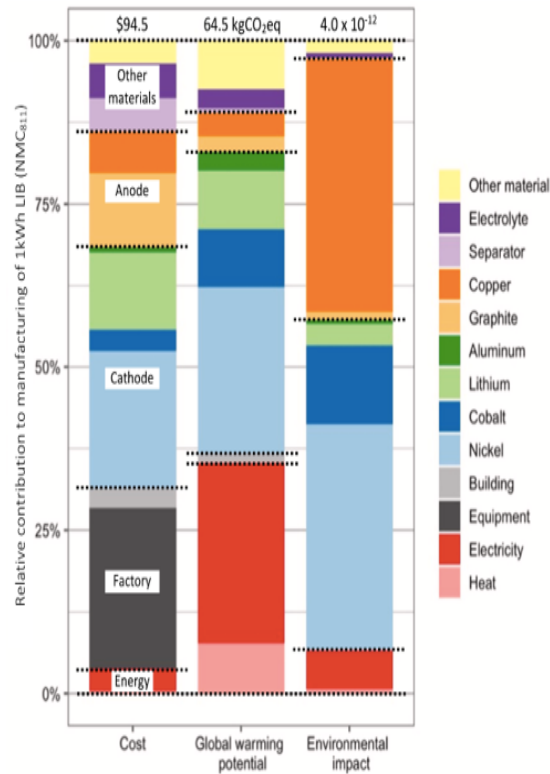


Figure 4 Environmental impact of lithium-ion battery (Gutsch and Leker 2024)

Table 7
required quantity of each material to produce 100 KJ.

Materials	Magnesium Nitrate Hexahydrate	Octadecane	Coconut oil using biofertilizers	Coconut oil using manurial fertilizers
Quantity (Kg)	0.794 Kg	0.4098 Kg	1.428 Kg	1.428 Kg

Table 8
ECI of each material to store/release 100 KJ.

Materials	Magnesium Nitrate Hexahydrate	Octadecane	Coconut oil using biofertilizers	Coconut oil using manurial fertilizers
ECI (£)	4.5	4.7	0.085	8.28

Figure 5 Environmental cost indicator of 4 PCMs and the required mass (Aridi and Yehya 2022)

2.3.2 Cost Analysis

As depicted in Figure 4, the cost of manufacturing a lithium battery (1kWh) for the energy required in our refrigerator would be approximately 190€. The average cost of PCMs ranges from €10 to €50 per kWh. This implies an additional maximum cost of 100€ per unit for a PV-driven fridge (Purohit and Dasgupta 2019). Since PCMs were utilized, there was no need for a battery. Consequently, it is less expensive than a commercialized PV-driven refrigerator with a battery.

3 Conclusions and recommendations

This report examines the possibility of using PCMs in a fridge to provide an alternative sustainable refrigeration method for regions without adequate electricity access. The PCM PlusICE A4 was selected based on specific criteria and integrated into the fridge. The PCM was placed in packs and these packs were distributed inside the fridge cabin with a 3kg mass constraint. An environmental impact and cost comparison with PV-driven fridges that use batteries showed that PCM fridges are more ecological and a lot more affordable.

The mass constraint makes the use of this technology challenging in areas with less sunlight, where the PCMs would need to be active for a longer period as the PV component would generate less current. Systems that use batteries don't experience this problem. To add on, temperature is more stable in battery using systems.

PCMs fridges are a promising new technology. However, its implementation is contingent on substantial solar energy availability, a factor that varies significantly based on geographical location. PCM fridges can still play an important role in optimizing energy use with their ability to store and release energy while being a way more ecological friendly solution than their battery-based counterpart. Further research should also be conducted addressing the possibility of combining PCMs with systems using batteries, maximizing the benefits of both energy storage solutions.

4 Annotated Reference List

- Aridi, R., and A. Yehya. 2022. "Review on the Sustainability of Phase-Change Materials Used in Buildings." *Energy Conversion and Management: X* 15 (August): 100237.
<https://doi.org/10.1016/J.ECMX.2022.100237>.
- Fleischer, Amy S. 2015. "Thermal Energy Storage Using Phase Change Materials: Fundamentals and Applications." *SpringerBriefs in Applied Sciences and Technology* 0 (9783319209210).
<https://doi.org/10.1007/978-3-319-20922-7>.
- Gutsch, Moritz, and Jens Leker. 2023. "Costs, Carbon Footprint, and Environmental Impacts of Lithium-Ion Batteries – From Cathode Active Material Synthesis to Cell Manufacturing and Recycling." *Applied Energy* 353 (January): 122132. <https://doi.org/10.1016/J.APENERGY.2023.122132>.
- Hillege L. 2023. "Environmental Cost Indicator (ECI) – Overview" accessed 10 December 2023, <<https://ecochain.com/blog/environmental-cost-indicator-eci/>>
- Khan, Mohammed Mumtaz A., R. Saidur, and Fahad A. Al-Sulaiman. 2017. "A Review for Phase Change Materials (PCMs) in Solar Absorption Refrigeration Systems." *Renewable and Sustainable Energy Reviews* 76 (September): 105–37. <https://doi.org/10.1016/j.rser.2017.03.070>.
- Purohit, Nilesh, and Mani Sankar Dasgupta. 2019. "Thermal Storage Material Enhanced Refrigerated Display Cabinet." In *Materials Today: Proceedings*. Vol. 28.
<https://doi.org/10.1016/j.matpr.2019.12.210>.
- Radhi, Mahdi M, J F Johain, and Abdul-Hadi N Khalifa. 2021. "Experimental Study of Cold Storage Packed Domestic Refrigerator with Solar Powered Variable Speed Compressor." *IOP Conference Series: Materials Science and Engineering* 1105 (1): 012060. <https://doi.org/10.1088/1757-899X/1105/1/012060>.
- Riffat, James, Cagri Kutlu, Emmanuel Tapia-Brito, Samuel Tekpetey, Francis B. Agyenim, Yuehong Su, and Saffa Riffat. 2022. "Development and Testing of a PCM Enhanced Domestic Refrigerator with Use

of Miniature DC Compressor for Weak/off Grid Locations.” *International Journal of Green Energy* 19 (10). <https://doi.org/10.1080/15435075.2021.1984244>.

1. Aridi: this work serves as a valuable resource for analyzing the environmental impact of PCM materials, particularly in the context of buildings. Notably, it investigates a PCM variant that aligns with the one employed in our research, enhancing the relevance and applicability of the findings to our specific investigation.
2. Fleischer: This book explains what a PCM is and the science behind its use for thermal storage applications like a fridge.
3. Gutsch: This reference is instrumental in understanding the carbon footprint and production impact of lithium-ion batteries during power outages. Through a comprehensive life-cycle analysis, this source not only enriches our understanding of lithium batteries but also sheds light on potential alternatives such as PCMs, aligning well with the goals of our research.
4. Khan: This scientific article was selected due to its comprehensive review of phase change materials (PCMs) in solar absorption refrigeration systems. It significantly contributes to our research theme on efficient thermal energy storage, aligning with the need for continuous operation. The content provides crucial insights into material choices and system performance.
5. Purohit: Purohit's research offers valuable insights into PCM applications in refrigerator cooling systems and provides cost data for raw PCM materials. This information is essential for comparing costs with lithium batteries, contributing significantly to the economic dimension of our research.

6. Radhi: This scientific source was chosen for its detailed examination of a solar-powered refrigerator with a variable speed compressor, utilizing phase change materials for thermal energy storage. The study provides valuable insights into the integration and performance of the system, aligning with research interests in renewable energy applications and sustainable cooling technologies.
7. Riffat: This article of a scientific journal studies the use of PCMs to develop a fridge for off-grid cooling (PCM selection and integration)